```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os
import zipfile
```

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.preprocessing import image
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout, BatchNormalization
from tensorflow.keras.applications import MobileNetV2
from sklearn.metrics import classification_report, confusion_matrix
```

```
import warnings
warnings.filterwarnings("ignore")
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import zipfile

with zipfile.ZipFile("/content/pet_dataset.zip", "r") as zip_ref:
    zip_ref.extractall("/content")

print("Dataset extracted successfully!")
```

```
Dataset extracted successfully!
```

```
import os
print(os.listdir("/content"))
```

```
['.config', '.ipynb_checkpoints', 'pet_dataset.zip', 'pet_dataset', 'drive', 'sample_data']
```

```
print(os.listdir("/content/pet_dataset"))
```

```
['test', 'val', 'train']
```

```
import os
dataset_path = "/content/pet_dataset"
print("Dataset Folder:")
print(os.listdir(dataset_path))
```

```
Dataset Folder:
['test', 'val', 'train']
```

```
train_path = "/content/pet_dataset/train"
print("Training Classes:")
print(os.listdir(train_path))
```

```
Training Classes:
['dogs', 'ferrets', 'hamsters', 'rabbits', 'cats']
```

```
test_path = "/content/pet_dataset/test"
print("Testing Classes:")
print(os.listdir(test_path))
```

```
Testing Classes:
['dogs', 'ferrets', 'hamsters', 'rabbits', 'cats']
```

```
import os

train_path = "/content/pet_dataset/train"

print("Training Dataset Summary")
print("-" * 30)
```

```
total_train = 0

for class_name in sorted(os.listdir(train_path)):

    class_path = os.path.join(train_path, class_name)

    if os.path.isdir(class_path):
        image_count = len(os.listdir(class_path))
        total_train += image_count
        print(f"{class_name}: {image_count} images")

print("-" * 30)
print(f"Total Training Images: {total_train}")
```

```
Training Dataset Summary
------------------------------
cats: 70 images
dogs: 70 images
ferrets: 70 images
hamsters: 70 images
rabbits: 70 images
------------------------------
Total Training Images: 350
```

```
import os

test_path = "/content/pet_dataset/test"

print("Testing Dataset Summary")
print("-" * 30)

total_test = 0

for class_name in sorted(os.listdir(test_path)):

    class_path = os.path.join(test_path, class_name)

    if os.path.isdir(class_path):
        image_count = len(os.listdir(class_path))
        total_test += image_count
        print(f"{class_name}: {image_count} images")

print("-" * 30)
print(f"Total Testing Images: {total_test}")
```

```
Testing Dataset Summary
------------------------------
cats: 15 images
dogs: 15 images
ferrets: 15 images
hamsters: 15 images
rabbits: 15 images
------------------------------
Total Testing Images: 75
```

```
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import os
```

```
train_path = "/content/pet_dataset/train"

classes = sorted(os.listdir(train_path))

plt.figure(figsize=(18, 6))

for i, class_name in enumerate(classes):

    class_folder = os.path.join(train_path, class_name)

    image_name = os.listdir(class_folder)[0]

    image_path = os.path.join(class_folder, image_name)

    img = mpimg.imread(image_path)

    plt.subplot(1, len(classes), i + 1)
```

```
    plt.imshow(img)

    plt.title(class_name.capitalize())

    plt.axis("off")

plt.tight_layout()
plt.show()
```


```
from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

```
train_datagen = ImageDataGenerator(
    rescale=1./255
)

test_datagen = ImageDataGenerator(
    rescale=1./255
)
```

```
train_generator = train_datagen.flow_from_directory(
    "/content/pet_dataset/train",
    target_size=(224, 224),
    batch_size=32,
    class_mode="categorical",
    shuffle=True
)
```

```
Found 350 images belonging to 5 classes.
```

```
test_generator = test_datagen.flow_from_directory(
    "/content/pet_dataset/test",
    target_size=(224, 224),
    batch_size=32,
    class_mode="categorical",
    shuffle=False
)
```

```
Found 75 images belonging to 5 classes.
```

```
from tensorflow.keras.applications import MobileNetV2
```

```
base_model = MobileNetV2(
    weights="imagenet",
    include_top=False,
    input_shape=(224, 224, 3)
)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_orde
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 2s 0us/step
```

```
base_model.trainable = False
```

```
from tensorflow.keras.models import Model
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
```

```
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(256, activation="relu")(x)
x = Dropout(0.5)(x)
predictions = Dense(5, activation="softmax")(x)
model = Model(inputs=base_model.input, outputs=predictions)
```

```
model.summary()
```

expanded_conv_proj… (BatchNormalizatio…	(None, 112, 112, 16)	64	expanded_conv_pr…
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_pr…
block_1_expand_BN (BatchNormalizatio…	(None, 112, 112, 96)	384	block_1_expand[0…
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_B…
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_r…
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]
block_1_depthwise_… (BatchNormalizatio…	(None, 56, 56, 96)	384	block_1_depthwis…
block_1_depthwise_… (ReLU)	(None, 56, 56, 96)	0	block_1_depthwis…
block_1_project (Conv2D)	(None, 56, 56, 24)	2,304	block_1_depthwis…
block_1_project_BN (BatchNormalizatio…	(None, 56, 56, 24)	96	block_1_project[…
block_2_expand (Conv2D)	(None, 56, 56, 144)	3,456	block_1_project_…
block_2_expand_BN (BatchNormalizatio…	(None, 56, 56, 144)	576	block_2_expand[0…
block_2_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_2_expand_B…
block_2_depthwise (DepthwiseConv2D)	(None, 56, 56, 144)	1,296	block_2_expand_r…
block_2_depthwise_… (BatchNormalizatio…	(None, 56, 56, 144)	576	block_2_depthwis…
block_2_depthwise_… (ReLU)	(None, 56, 56, 144)	0	block_2_depthwis…
block_2_project (Conv2D)	(None, 56, 56, 24)	3,456	block_2_depthwis…
block_2_project_BN (BatchNormalizatio…	(None, 56, 56, 24)	96	block_2_project[…
block_2_add (Add)	(None, 56, 56, 24)	0	block_1_project_… block_2_project_…
block_3_expand	(None, 56, 56,	3,456	block_2_add[0][0]

(Conv2D)	144)		
block_3_expand_BN (BatchNormalizatio…	(None, 56, 56, 144)	576	block_3_expand[0…
block_3_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_3_expand_B…
block_3_pad (ZeroPadding2D)	(None, 57, 57, 144)	0	block_3_expand_r…
block_3_depthwise (DepthwiseConv2D)	(None, 28, 28, 144)	1,296	block_3_pad[0][0]
block_3_depthwise_… (BatchNormalizatio…	(None, 28, 28, 144)	576	block_3_depthwis…
block_3_depthwise_… (ReLU)	(None, 28, 28, 144)	0	block_3_depthwis…
block_3_project (Conv2D)	(None, 28, 28, 32)	4,608	block_3_depthwis…
block_3_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_3_project[…
block_4_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_3_project_…
block_4_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_4_expand[0…
block_4_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_4_expand_B…
block_4_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_4_expand_r…
block_4_depthwise_… (BatchNormalizatio…	(None, 28, 28, 192)	768	block_4_depthwis…
block_4_depthwise_… (ReLU)	(None, 28, 28, 192)	0	block_4_depthwis…
block_4_project (Conv2D)	(None, 28, 28, 32)	6,144	block_4_depthwis…
block_4_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_4_project[…
block_4_add (Add)	(None, 28, 28, 32)	0	block_3_project_… block_4_project_…
block_5_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_4_add[0][0]
block_5_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_5_expand[0…
block_5_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_5_expand_B…
block_5_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_5_expand_r…
block_5_depthwise_… (BatchNormalizatio…	(None, 28, 28, 192)	768	block_5_depthwis…
block_5_depthwise_… (ReLU)	(None, 28, 28, 192)	0	block_5_depthwis…
block_5_project (Conv2D)	(None, 28, 28, 32)	6,144	block_5_depthwis…
block_5_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_5_project[…
block_5_add (Add)	(None, 28, 28, 32)	0	block_4_add[0][0… block_5_project_…
block_6_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_5_add[0][0]
block_6_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_6_expand[0…
block_6_expand_relu	(None, 28, 28,	0	block_6_expand_B…